Identifying the quotient flag graphs

Human Compression project

This note completes the finite calculation for the first atlas in Hartley, Hubard and Leemans, *Two atlases of abstract chiral polytopes for small groups*, Ars Math. Contemp. 5 (2012), 371–382. The second atlas, for almost simple automorphism groups, is a separate outstanding calculation. We acquire the earlier complete atlas of regular covers, the reviewed normalizer criterion and marked-cover obstruction, and the explicit verification of the 56 positive examples. In particular, the completeness of the earlier regular atlas is a retained input. Current website files are bound literally; historical byte identity is not asserted.

Let $W = \langle r_0, \ldots, r_{d-1} \rangle$ be a supplied marked permutation group and let $N \leq W$ be one of its retained flag stabilizers. All cores are allowed. The first calculation supplies $|W|$, the generators of N, its order, and the conditions on its normalizer. Its subgroup enumeration uses the stated GAP software premises. The following additional check uses only literal permutations and finite graphs.

Proposition 1 (Checking a coset graph)**.** *Let X be a connected graph with one adjacency $x \mapsto x^i$ of each colour i at each vertex, and choose $x_0 \in X$. Choose permutations $g_x \in W$ by following a breadth-first traversal from $g_{x_0} = 1$. Suppose that*

$$|X|\,|N| = |W|, \qquad g_x r_i g_{x^i}^{-1} \in N \quad (x \in X,\ 0 \leq i < d).$$

Then $x \mapsto Ng_x$ is an isomorphism from the coloured graph X to the right coset graph of N in W with the given generators.

Proof. The displayed membership is exactly the equality $Ng_x r_i = Ng_{x^i}$. Thus the map preserves every adjacency and sends x_0 to N. Every element of W is a word in the marked generators. Following that word from x_0 shows that every coset occurs in the image. The cardinality equality makes this surjective map bijective. □

The checker enumerates N from its literal generators and verifies its claimed order. It constructs every g_x as a word in the original marks and checks every displayed equality. The value of $|W|$ is acquired from the preceding exact group calculation. An abstract database identifier is not used in this argument.

Proposition 2 (Reusing a verified flag graph)**.** *In a connected graph with specified colours and one adjacency of each colour, an isomorphism is determined by the image of one vertex. Label the vertices in breadth-first order, always exploring colours in their specified order, and record the resulting adjacency array. Two rooted graphs have the same array if and only if the traversal orders give a colour-preserving isomorphism between them.*

Proof. An isomorphism must send each path from the root to the path with the same colours from its image. Consequently it preserves the traversal order and recorded array. Conversely equality of the arrays makes the bijection between the two traversal orders preserve every coloured edge. □

Each of the 56 acquired chiral flag graphs has exactly two automorphism orbits, represented by a chosen vertex and one of its adjacent vertices. The 112 corresponding arrays therefore suffice for comparison with every possible root image. Every successful comparison also supplies

the full vertex bijection, and every edge of that bijection is checked. The acquired polytope axioms, chirality and group identifications then transfer through this explicit isomorphism.

For an unmatched graph, omit colour i to obtain its prospective rank-i faces as connected components. In an actual polytope these are precisely the faces containing the flags in those components. Two particular necessary conditions suffice for all unmatched graphs in this calculation. First, each interval between ranks $i-1$ and $i+1$ has exactly two intermediate faces, with the least and greatest faces included at the boundary ranks. Second, flags containing the same fixed chain are connected by adjacencies that change none of those faces. The first condition is the diamond axiom; the second is strong flag connectedness. A failure is recorded by the actual face labels and representative flags, or by two flags in distinct allowed components with the same fixed faces. These are necessary conditions in every rank; passing them alone is not used as a positive certificate.

The complete calculation. The current regular-atlas input contains 9,212 nondegenerate markings. The earlier product argument excludes the markings with a bond labelled two; ranks one and two cannot be chiral. The three retained execution intervals cover respectively the source indices $[0, 118)$, $[118, 7302)$ and $[7302, 9212)$, without overlap. The first two interval receipts remain failed or interrupted; their individual completed entries are acquired from the exact completion journal and saved records. No interrupted entry contributes an exclusion.

The normalizer conditions yield 8,986 candidates across those markings. The base-face intersection test leaves 756. Literal coset transport checks 704,608 coloured edges of these quotient graphs. Of the 756, 360 have explicit isomorphisms to the acquired examples and cover all 56 of those examples. The other 396 have recorded obstructions: 350 violate a diamond condition and 46 violate flag-residue connectivity. There are no undecided candidates. The verified examples are pairwise nonisomorphic as coloured flag graphs, so this gives exactly 48 of rank three and eight of rank four, with a distinct dual counted separately.

Scope and attribution. The result is relative to the stated complete regular-cover input and exact group-arithmetic premises. The normalizer description and the original 56 examples retain their original attribution. The contribution here is a complete calculation with arbitrary action kernels and explicit transport to reusable verified examples, together with literal negative witnesses. No claim about the second atlas, a new open question, Lean formalization or measured compression follows from this component.